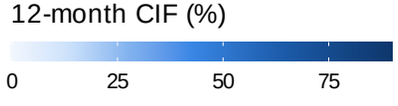
Supplementary: unadjusted 12-month cumulative incidence (Aalen–Johansen)

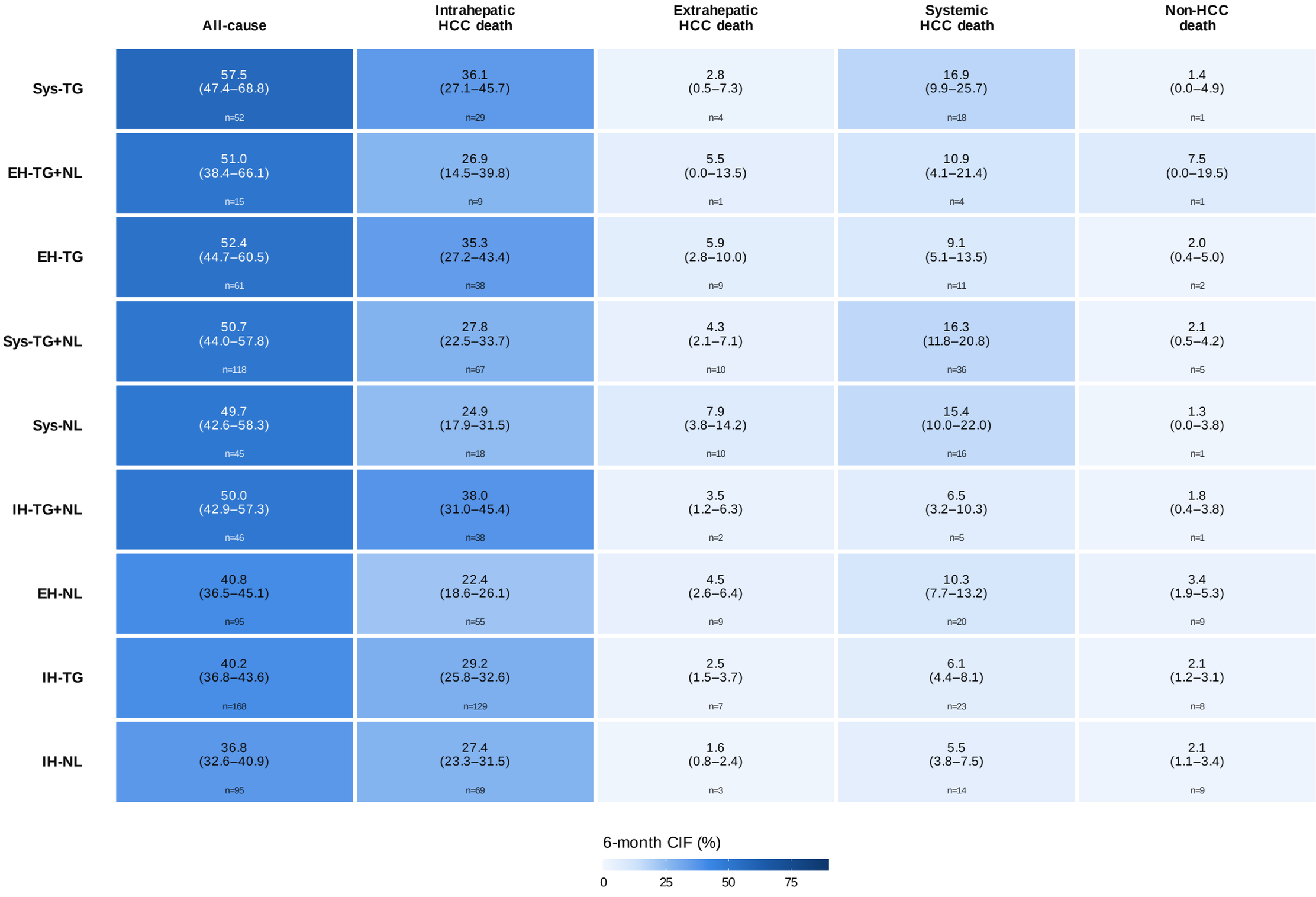
% (95% CI); all-cause = 1 − Kaplan–Meier; n = events within 12 months

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
Sys-TG	83.4 (70.5–90.7) n=55	47.5 (36.6–61.7) n=31	5.7 (2.2–14.8) n=4	28.6 (19.5–41.8) n=19	1.6 (0.2–11.2) n=1
EH-TG+NL	74.0 (51.2–86.2) n=21	35.0 (21.2–57.8) n=10	7.2 (1.9–27.3) n=2	20.9 (10.3–42.8) n=6	10.9 (3.7–31.7) n=3
EH-TG	63.4 (53.1–71.4) n=72	36.4 (28.6–46.4) n=42	10.0 (5.7–17.5) n=11	13.5 (8.4–21.6) n=15	2.7 (0.9–8.1) n=3
Sys-TG+NL	78.1 (70.7–83.7) n=135	43.3 (36.4–51.4) n=75	7.0 (4.1–12.1) n=12	24.3 (18.7–31.7) n=42	3.5 (1.6–7.7) n=6
Sys-NL	81.0 (69.2–88.2) n=63	33.0 (24.1–45.4) n=26	16.8 (10.2–27.7) n=13	28.5 (19.9–40.6) n=22	2.7 (0.7–10.6) n=2
IH-TG+NL	73.0 (62.3–80.7) n=74	53.8 (44.7–64.6) n=55	6.2 (2.9–13.6) n=6	9.9 (5.5–17.8) n=10	3.2 (1.0–9.7) n=3
EH-NL	62.9 (56.2–68.6) n=159	29.3 (24.2–35.5) n=76	7.3 (4.7–11.4) n=18	19.7 (15.3–25.5) n=48	5.9 (3.6–9.6) n=15
IH-TG	66.8 (61.7–71.3) n=269	47.4 (42.7–52.6) n=193	5.0 (3.2–7.7) n=19	9.6 (7.1–13.1) n=38	4.0 (2.5–6.5) n=16
IH-NL	49.8 (43.9–55.1) n=161	33.4 (28.6–39.0) n=109	2.6 (1.3–5.1) n=8	9.2 (6.5–13.0) n=29	4.7 (2.8–7.7) n=15



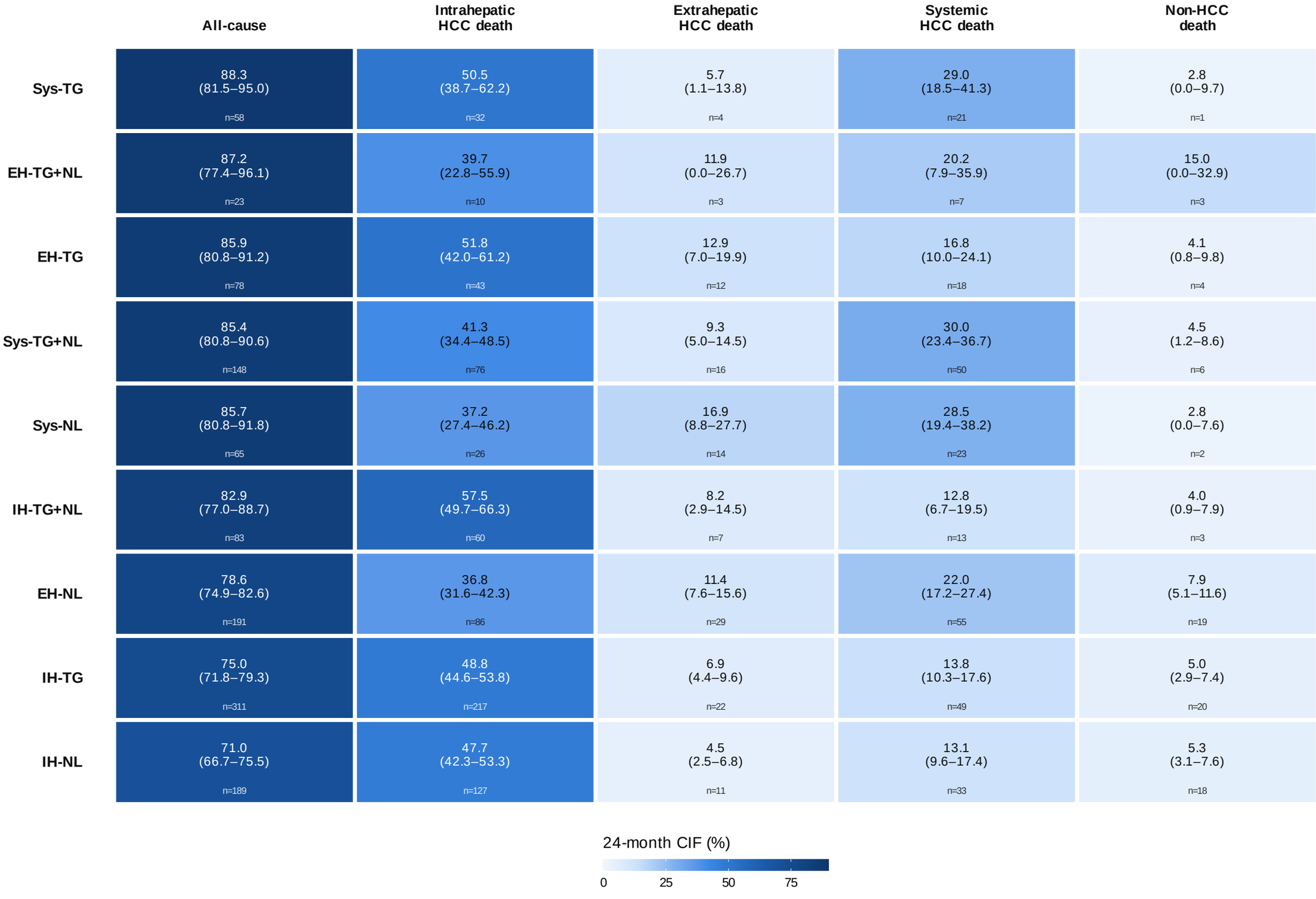
Supplementary: adjusted 6-month CIF after first progression

Same models as Figure 4.4A; n = events within 6 months



Supplementary: adjusted 24-month CIF after first progression

Same models as Figure 4.4A; n = events within 24 months



A2 Absolute risk difference at 6 months vs population-standardised average

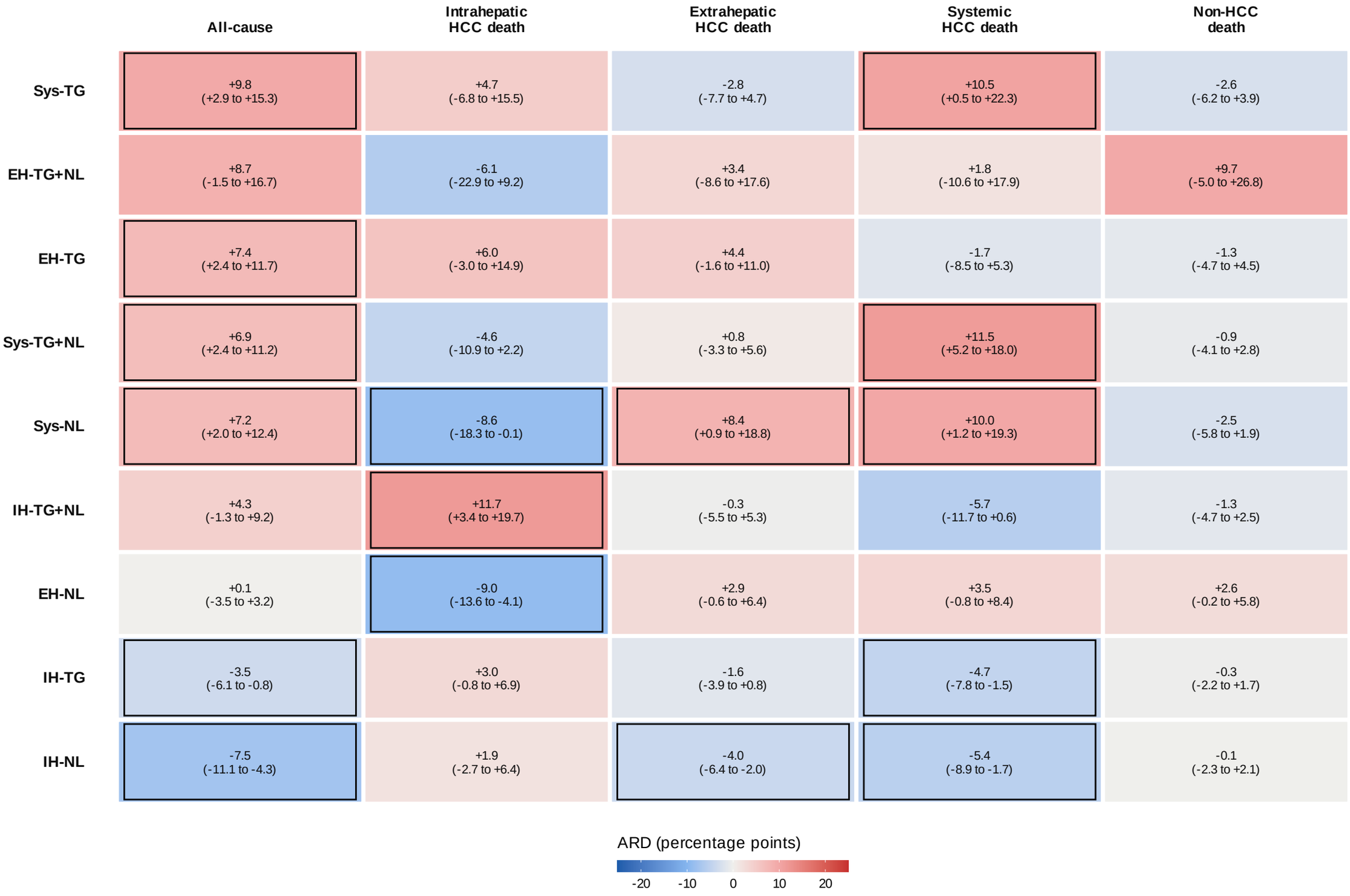
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

A2 Absolute risk difference at 24 months vs population-standardised average

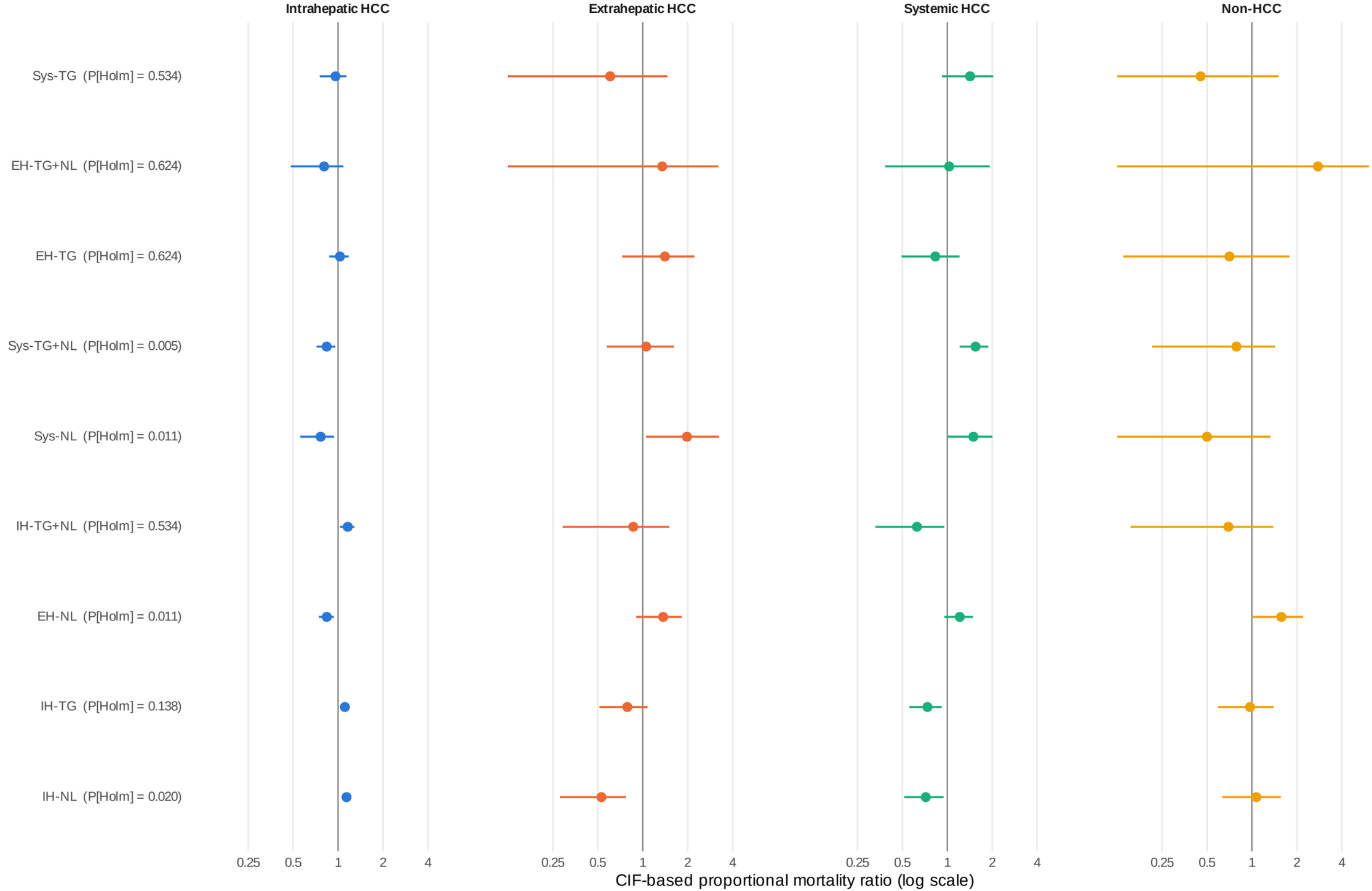
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

A4 Cause-of-death enrichment: CIF-based proportional mortality ratio (PMR) at 6 months

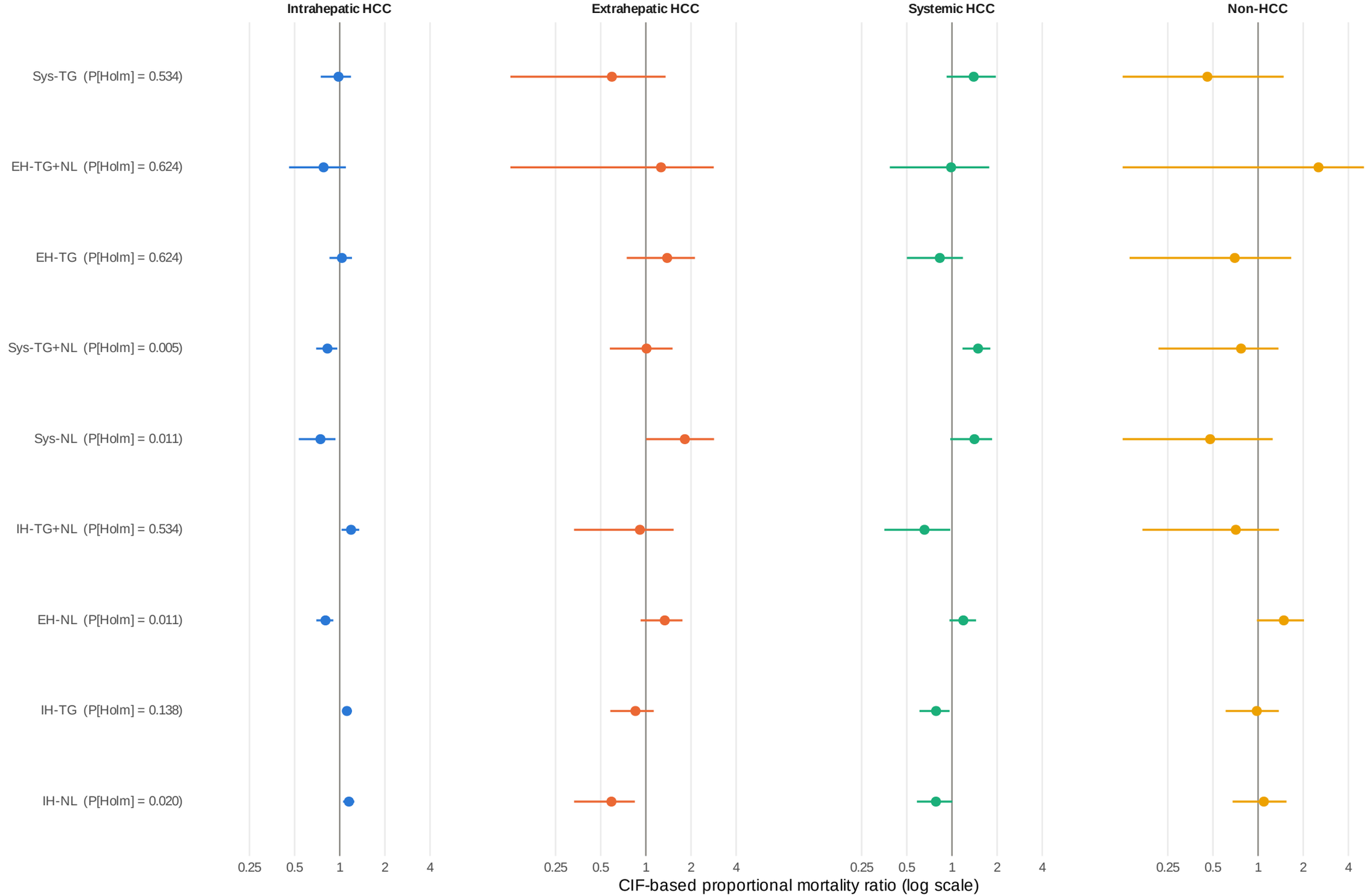
PMR = share of deaths from cause j within the pattern / share in all progressors. 95% bootstrap CI (clipped at 1/8 and 8).
P[Holm]: weighted-effect-coded Wald test (df = 2) that the pattern's csHR is equal across IH/EH/Systemic HCC death (from Lunn–McNeil model).



PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

A4 Cause-of-death enrichment: CIF-based proportional mortality ratio (PMR) at 24 months

PMR = share of deaths from cause j within the pattern / share in all progressors. 95% bootstrap CI (clipped at 1/8 and 8).
P[Holm]: weighted-effect-coded Wald test (df = 2) that the pattern's csHR is equal across IH/EH/Systemic HCC death (from Lunn–McNeil model).

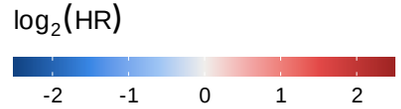


PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

Supplementary: Firth-penalised HR / csHR (reference: Sys-TG)

coxphf (Firth-penalised partial likelihood, Wald CI from penalised information); treatment group entered as covariate (coxphf has no strata)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.53 (0.40–0.70)***	0.61 (0.42–0.89)**	0.68 (0.23–1.95)	0.29 (0.17–0.49)***	NE (sparse reference)
IH-NL	0.45 (0.33–0.61)***	0.55 (0.37–0.81)**	0.40 (0.13–1.21)	0.24 (0.14–0.43)***	NE (sparse reference)
IH-TG+NL	0.70 (0.50–0.98)*	0.87 (0.56–1.35)	1.03 (0.30–3.51)	0.36 (0.18–0.72)**	NE (sparse reference)
EH-TG	0.82 (0.58–1.15)	0.85 (0.54–1.35)	1.80 (0.59–5.45)	0.50 (0.26–0.95)*	NE (sparse reference)
EH-NL	0.57 (0.42–0.77)***	0.46 (0.31–0.70)***	1.23 (0.43–3.48)	0.48 (0.28–0.80)**	NE (sparse reference)
EH-TG+NL	0.80 (0.49–1.31)	0.68 (0.33–1.38)	1.94 (0.44–8.55)	0.58 (0.24–1.39)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.79 (0.55–1.13)	0.59 (0.35–0.99)*	2.48 (0.81–7.63)	0.82 (0.44–1.50)	NE (sparse reference)
Sys-TG+NL	0.78 (0.57–1.06)	0.67 (0.44–1.02)	1.24 (0.42–3.68)	0.86 (0.51–1.45)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.310**

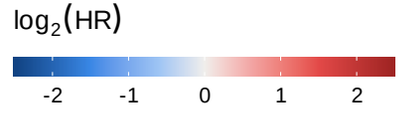


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR excluding undetermined deaths (reference: Sys-TG)

n = 1778 (8 deaths with unclassified/unknown/missing cause removed)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.54 (0.40–0.71)***	0.64 (0.44–0.93)*	0.77 (0.26–2.28)	0.28 (0.17–0.48)***	NE (sparse reference)
IH-NL	0.46 (0.34–0.62)***	0.58 (0.39–0.86)**	0.46 (0.15–1.42)	0.24 (0.14–0.43)***	NE (sparse reference)
IH-TG+NL	0.73 (0.52–1.02)	0.94 (0.61–1.46)	1.14 (0.32–4.00)	0.34 (0.16–0.68)**	NE (sparse reference)
EH-TG	0.83 (0.59–1.17)	0.90 (0.56–1.43)	2.02 (0.65–6.28)	0.48 (0.25–0.92)*	NE (sparse reference)
EH-NL	0.57 (0.42–0.77)***	0.48 (0.32–0.72)***	1.36 (0.46–3.96)	0.47 (0.28–0.79)**	NE (sparse reference)
EH-TG+NL	0.79 (0.48–1.29)	0.65 (0.32–1.33)	1.84 (0.39–8.68)	0.56 (0.23–1.36)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.78 (0.55–1.12)	0.59 (0.35–0.99)*	2.64 (0.83–8.34)	0.79 (0.43–1.46)	NE (sparse reference)
Sys-TG+NL	0.79 (0.58–1.07)	0.68 (0.45–1.03)	1.43 (0.47–4.36)	0.86 (0.51–1.45)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.411**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR including patients flagged for deletion (reference: Sys-TG)

n = 1797



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.